# LECTURE 5

Welcome to lecture 5. The term evaluation is normally confused with assessment in most cases, in this particular lesson we will endeavor to clarify this and ensure that the meaning is brought out clearly. The purpose of evaluation will also be elucidated together with the different types of evaluation techniques. Let us first look at the learning outcomes for the lesson.

## EVALUATION AND PERFORMANCE STANDARDS

### Learning objectives

After this lecture, the learner should be able to:

**a.** Define evaluation
b. Discuss the purpose of evaluation.
c. Discuss types of educational evaluation.
d. Discuss standards of evaluation

## What is evaluation?

In terms of measurement evaluation is defined as ***the ability to make judgment on the basis of given information. Judging the worthiness of a programme project or course.*** It is the use of measurement in making decisions.

## The Birth of Evaluation

**STORY 1: A story about the "birth" of evaluation" (They story of creation and the concept of Evaluation)**

Genesis Chapter 1: In the beginning God created the heavens and the earth'
Genesis Chapter 31: God saw all that he had done, and it was very good.

A story is told to illustrate the birth or the concept of evaluation. "In the beginning God created the heavens and the earth (Genesis Chapter 31). And God saw everything that He made, and said "behold, it is very good". And the evening and the morning were created on the sixth day. And on the seventh day God rested from all His work. His archangel came then to Him asking "God, how do you know that what you have created is "very good?" What are your criteria? On what

data do you base your judgement? Aren't you a little too close to the situation to make a fair and unbiased evaluation? God thought about these questions all day and His rest was greatly disturbed. On the eight-day God said, "Lucifer, go to hell". Thus was evaluation born in a blaze of glory …. a legacy under which we continue to operate.

**STORY 2: A story about what evaluation is about: The Story of lost key and the concept of Evaluation**

Evaluation is about making judgement and asking questions. A story is told of a man who once found his neighbor down on his knees under a street lamp looking for something.

"What have you lost my friend?"

"My key", said the man on his knees.

After minutes of searching the neighbor asked,

"Where did you drop it?"

"In that dark pasture" replied his friend.

"Then why for heaven's sake are you looking or it here?"

"Because there is more light here".

The foregoing story serves to illustrate what evaluation is out to achieve. ***Evaluation is simply an attempt to throw some light on the dark parts of a project or programme.***

## SUBJECTS FOR EVALUATION: WHAT DO WE EVALUATE?

There are many areas that can be evaluated including:

1. ***Students***

For students' evaluation, we cover such variable as:

- Achievement
- Aptitude
- Intelligence
- Personality
- Attitudes

2. ***Curriculum***

For curriculum evaluation, we cover such areas as:

- Instructional materials
- Instructional strategies
- Textbooks
- Audio-visual materials
- Organisation

3. ***School***

For school evaluation, we focus on how education programme with respect to its functioning:

- School activities
- School resources

4. ***Programme***

For programme evaluation, we look at the following:

- Needs assessment
- Initial objectives of the programme
- Activities
- Impact

5. ***Personnel***

For personnel evaluation (staff evaluation) we look at:

- Performance
- Productivity

## Evaluation Design

Assignment:

- Evaluate the teaching of national and cohesion at the university
- Evaluate school feeding programme in ASAL areas

To do these exercises you need a design.

A design is a road map, an architect or plan of doing something. How do you go about doing it? Under evaluation design, you need to discuss:

- Purpose of the evaluation-why carry out evaluation?
- Evaluation objects (areas and questions)
- Evaluation model or methodology

***Evaluation Objects: Questions you ask when carrying out an evaluation of a Project***

What major areas, also known as **evaluation objects,** do you focus on when you are carrying out an end of the project evaluation exercise? What questions do you ask when you are carrying out an evaluation? An evaluator is seeking answers for the following questions:

- What were the intentions/**objectives** of the programme/project? Examine objectives.
- What have been achieved (**the results)?** Examine the results.
- What is still there **(impact)?** Examine the impact (what is still there on the ground).
- What is the future of the project **(sustainability)?**

***The Evaluation Model/Methodology***

The design of the study is to a large extent set by the frames of the assignment. The data has to be collected.

Methodology refers to how you collect data for evaluating a programme/project. Evaluation leans on collection of both qualitative and quantitative data. The instruments you can use to collect evaluation data include:

- *Review of Documents/ reports* and policy documents. These provide questions to be asked in either the interview or questionnaire.

- *Interview with stakeholders*/ actually participants in the project/key players.
- *Observations.* If it was classroom-based project e.g. teaching programme, carry out classroom observations.
- *A questionnaire*. This is used to tap some quantitative data regarding the programme/project.

# Purposes of Evaluation

*Why carry out an Evaluation? / What is the purpose of evaluation?*

Evaluations are initiated for many reasons, including: -

i. ***To assess programme improvement***
ii. ***To assess accountability***
iii. ***To assess an educational process [Tracking Students Progress]***
iv. ***To assess achievement of interventions and for knowledge generation***
v. ***For hidden agendas***

## 1. Programme Improvement

This is an evaluation that is intended to provide information for guiding programme improvements. The purpose for this evaluation is to help or shape the programme to perform better.

## 2. Accountability [e.g. Evaluating School Feeding Programme]

This is an evaluation conducted to determine whether the project expectations have been met. The purpose for this evaluation is to render a summary judgment on the programmes performance.

***Example***

The investment in school feeding programme is justified by the presumption that the programme will attract children to school or benefit children in those areas. Program managers are thus expected to use resources effectively and efficiently and actually produce the intended benefits.

First examine the objectives of the feeding programme – why was the feeding programme introduced?

- ***Objective:*** To assist the children from the poor background to attend school.
- ***Question/results:*** Has higher school attendance been achieved? Are there now more children from poor background in school than before the project was introduced?
    - If yes, then the project has met the objectives.
    - If no, then the project has failed.

Is there still higher school attendance since the project ended? (Impact).

How is high attendance that has been achieved going to be maintained in the future? (Sustainability).

Evaluating a course e.g. geography. First state the objectives of why geography should be taught in high school. Ask the learners geographical facts to evaluate the achievement / failure of geography.

## 3. Evaluation a tool for Tracking Students Progress

A primary purpose of teaching is to produce a measurable change in behavior. Evaluation as an educational process is an assessment on how well learners are doing and what effect the project is having on them.

| Student | Test 1 | Test 2 | Test 3 | Average |
|---|---|---|---|---|
| **Student A** | 6 | 6 | 6 | 6 |
| **Student B** | 4 | 6 | 8 | 6 |

Here you can see that both students have an average grade of 6. But what does it really say? If you just show your student this grade, it means nothing. If you dig deeper and take a look at the process,

you can see that one student actually did better than the other. Student B shows progress and improvement in the learning material. Student A has the same grade as Student B, but he's stuck. He doesn't get a complete grip on the learning material, and only masters some parts of it.

## 4. Knowledge Generation through interventions

- Sometimes evaluations are undertaken to describe the nature and effects of an intervention as a contribution to knowledge.
- To determine the extent to which the goals of interventions have been achieved.

Intervention

Smoking Addiction Stop

| |
|---|
| Demonstration/ intervention program or on a new approach to a social problem e.g.Therapy on drug addiction such as giving them Coffee every one hour. |

Smoking

An evaluation can be carried out to find out if it can be implemented more widely.

## 5. Hidden Agendas

Sometimes the true purpose of the evaluation has little to do with actually obtaining information about the programme's performance.

i. Program administrators or boards may launch an evaluation because they believe it will be good public relations and might impress funders or political decision makers.

ii. Occasionally, an evaluation is commissioned to provide a rationale for a decision that has already been made behind the scenes to terminate a program, fire an administrator etc.

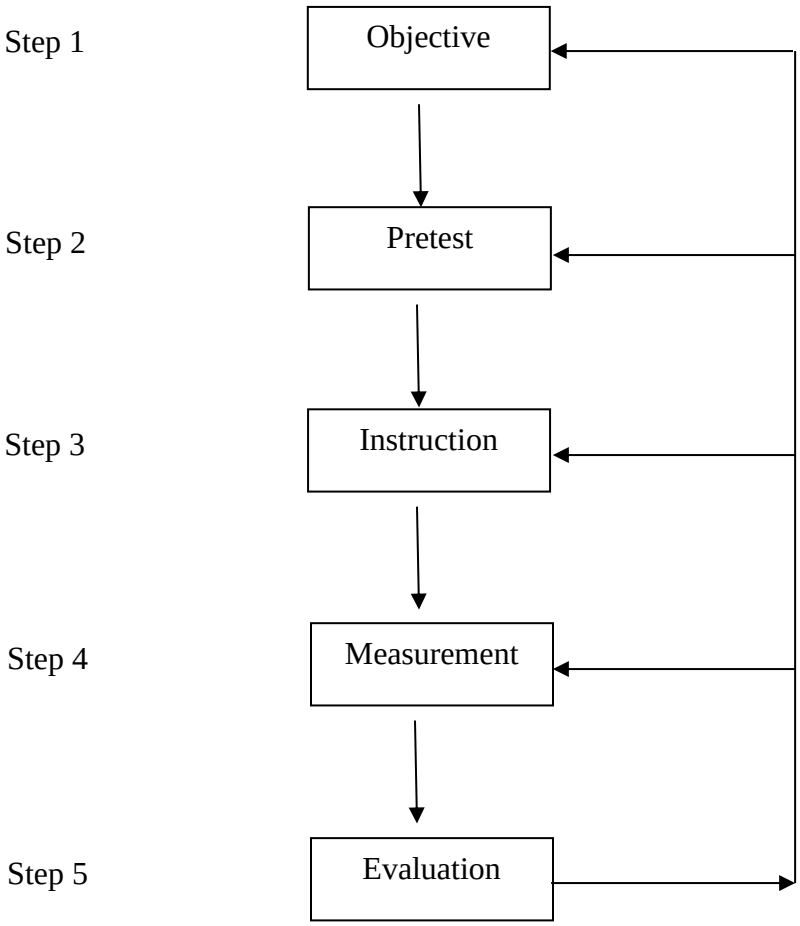
iii. The evaluation may be commissioned as a delaying tactic to appease critics and defer difficult decisions e.g. Salaries and Remuneration Commission job evaluation delaying tactic on teacher's salaries.

## EVALUATION PHASES AND PROJECT CYCLE

Evaluation is a continuous process. The evaluation process is cyclic with feedback from one cycle guiding the next. We do not just evaluate outcomes: every stage of the process is subject to evaluation beginning with the objectives.

***Example 1: Evaluation of student achievement***

A teacher uses performance data not only to evaluate student progress but also to evaluate his/her own instruction. That is, the process of evaluating students provides feedback to the teacher.
The components of the evaluation model are:



The figure given below is an evaluation model appropriate for use in primary and secondary educational settings.

***Step 1: Objective***

Preparation of the objectives is the first step in the evaluation process, because objectives determine what we will seek to achieve. The objectives give direction to instruction and define what behaviors we want to change.

***Step 2: Pretest***

With some type of pretest, we can answer three questions:

1. How much had already been learned?
2. What are the individual's current status and capabilities?
3. What type of activity should be prescribed to help achieve the objectives?

***Step 3: Instruction***

Sound instructional methods are needed to achieve the agreed-on objectives. Different instructional procedures may be needed to meet students' individual needs.

***Step 4: Measurement***

This involves the selection or development of a test to gauge the achievement of the objectives. It is crucial that the test be designed to measure the behavior specified in the objectives. The objectives can be cognitive, affective, or psychomotor. The key element is to select or develop a test that measures the objectives. Often, teachers will need to develop their own tests, because standardized tests may not be consistent with their instructional objectives. This is a common method used to provide evidence of test validity in educational settings.

***Step 5: Evaluation***

Once the instructional phase has been completed and achievement has been measured, test results are judged (i.e. evaluated) to find whether the desired changes achieved the stated objective.

What happens if students do not achieve the desire objectives? The figure given above shows a feedback loop from evaluation back to each component of the model. Failure to achieve the stated objectives may be due to any segment of the model. First, it may be discovered that the objectives are not appropriate and may need to be altered. The instruction may not have been suitable for the group, or the selected test may not have been appropriate. The educational evaluation model is dynamic. The evaluation model provides information needed to alter any aspect of the educational process.

***Example 2: Project Evaluation Phases***

Other situations where evaluation is applied include projects. The main stages that have been identified on project cycle are:

1. ***Project Identification*.** This involves:

- Baseline survey** [Situation analysis]
- Information / data gathering
- Prioritization of the challenges

2. ***Project Planning***. This involves:

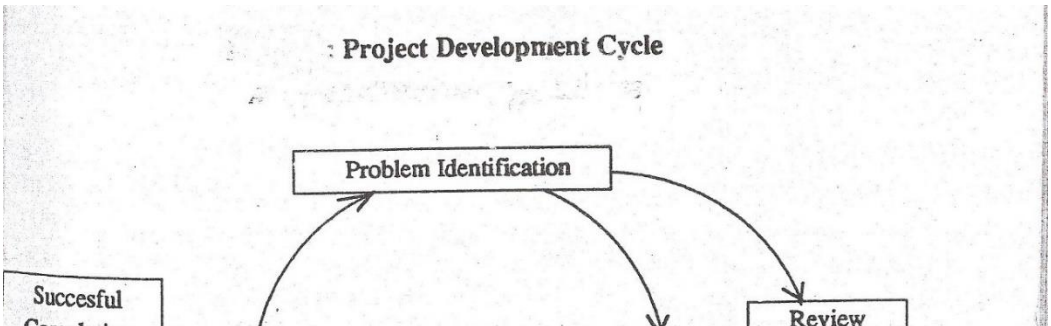
- Identification of resources
- Setting project objectives expressed in terms of quantifiable, measurable outcomes.
- Identification of risks and contingencies (E.g. political instability, donor withdrawal)

3. ***Project implementation***. This involves:

- Role division
- Team work
- Monitoring activities (periodic review of progress)
- Budget control

4. ***Project Evaluation*.** This involves:

- Effect / impact assessment
- Identifying side effects
- Accountability



Donors like the World Bank and African Development Bank use Project Cycle given above in evaluating the projects they fund.

**** Baseline survey** is the survey carried out before a project is initiated thus providing a basis for comparative judgment on the situation after intervention.

## *Project Risks*

Any project has risks that need to be pointed out. In any project we have:

***Manifest effects***= Intended effects e.g, encouraging school enrollment

***Latent effects***= Unintended effects and these include **project risks**. E.g warriors also became interested in going to school.

In Kenya project risks may include:

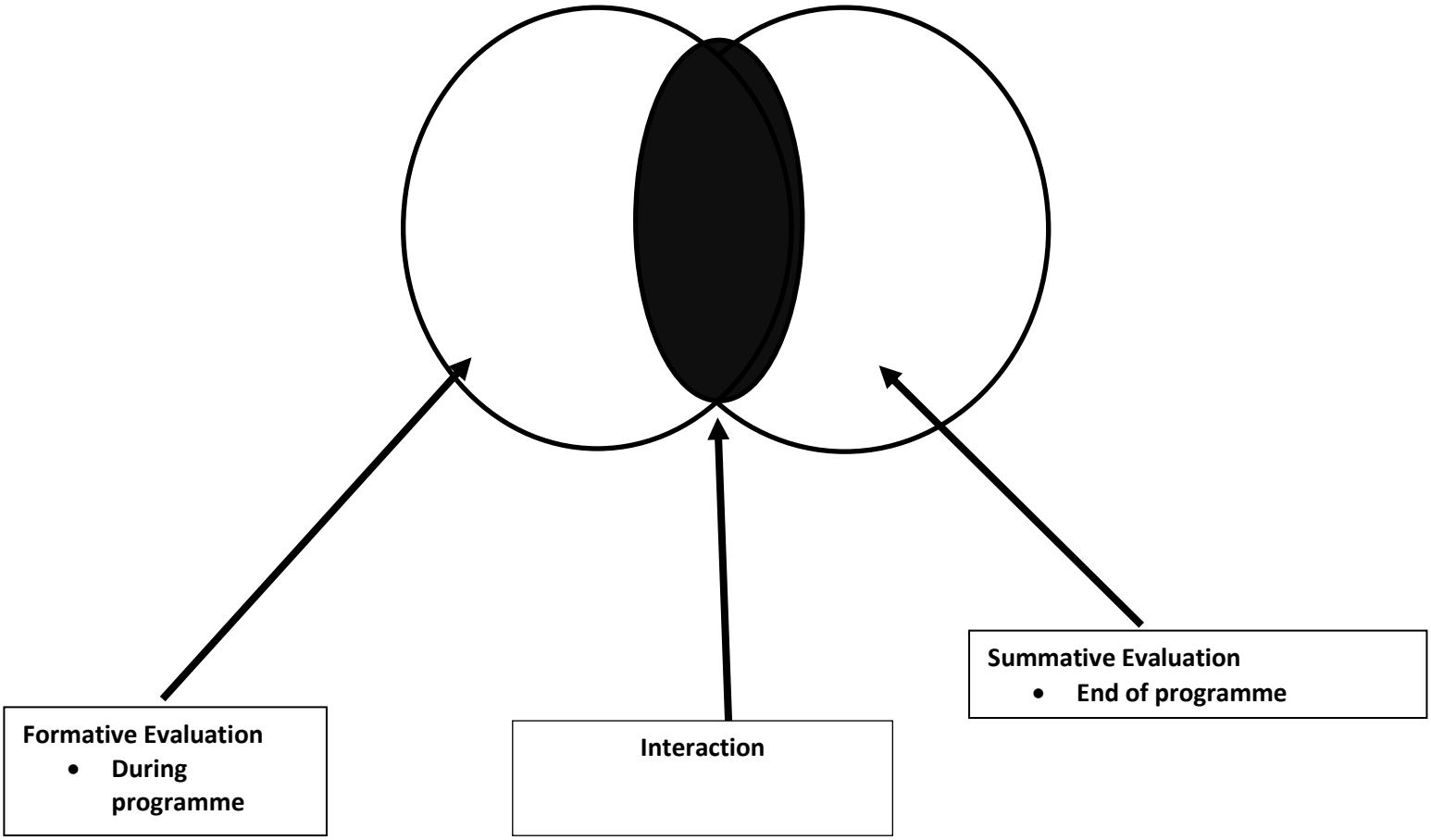
i. Political instability.
ii. Luck of funds to complete the project.
iii. Change of leadership.

The concepts of manifest and latent effects are sociological concepts first formulated by Robert Merton.

# TYPES OF EDUCATIONAL EVALUATION

There are two types of educational evaluation, namely:

- Formative evaluation
- Summative evaluation




## Formative Evaluation

This is the judgment of achievement during the formative stages of learning. Feedback is one of the most powerful variables in learning. Formative evaluation is used to provide feedback to learners throughout the instructional process.

The main purpose of formative evaluation is to determine the degree of mastery of a given learning task and to pinpoint the part of the task not mastered. The strength of formative evaluation is that it is used to provide feedback throughout the instructional unit.

### Summative Evaluation

This is the judgment of achievement at the end of an instructional unit, and typically involves the administration of tests at the conclusion of an instructional unit or training period. Summative means ***"totaling up"*** to indicate the level a learner has reached in a subject.

Similarities and Differences between Formative and Summative Evaluation

| | **Formative** | **Summative** |
|---|---|---|
| Purpose | Feedback to student and teacher on student progress throughout an instructional unit. | Certification or grading at the end of a unit, semester or course. |
| Time | During instruction. | At the end of a unit, semester, or course. |
| Emphasis in Evaluation | Explicitly defined behavior. | Broader categories of behaviors or combinations of several specific behaviors. |
| Performance Standard | Criterion – referenced. | Norm-referenced or criterion-referenced. |

# PERFORMANCE STANDARDS

Performance standards are the criteria to which the results of measurement are compared in order to interpret them. A test score in and of itself means nothing. There are two ways with which we can interpret the results of a test. The two most widely used types of performance standards are:

- Norm – referenced standards
- Criterion – referenced standards

These are the two bases for the comparison of performance or interpretation of performance.

## Norm – Referenced Standards

A norm–referenced standard or test is used to judge an individual's performance in relation to the performances of other members of a well-defined group – for example, 14 years old. KCPE is an example of a norm – referenced standard or test. That is, norm – referenced tests, yield information regarding the student's performance in comparison to a norm or average of performance by similar students.

Norm – referenced standards are developed by testing a large number of people of a defined group. Descriptive statistics are then used to develop standards. A common norming method is to use percentile ranks.

This is where comparison in performance is made between individuals. The comparison is governed by norms e.g. *they have all followed the same curriculum and are of the same age.*

In summary ***norm-referenced standard*** involves:

i. Comparing individual performance to the performance of the group.
ii. Grading on a normal curve. Norm-referenced evaluation test items are chosen to produce variability among test scores so that a normal curve can be derived.
iii. Plotting all scores for a group against the number of students who achieve each score and any individual's standing is interpreted according to its variation from the group norm.
iv. Used for determining an individual's relative position in a group.

## Criterion – Referenced Standards

A criterion-referenced standard or test is used to determine if someone has attained a specific level. [e. g. breaking world record in 100 metres]. CRS is also referred to as objective-referenced or competence-based evaluation.

A criterion-referenced standard is a predetermined standard of performance that shows the individual has achieved a desired level of performance. It is unlike a norm-referenced standard in that *performance of the individual is not compared with that of other individuals, but rather just against the standard.* That is:

- Criterion referenced tests provide information about a student's level of proficiency in or mastery of some skills or set of skills.

- It tells us whether a student needs more or less work on skills, but says nothing about the student's performance relative to other students.
- Grading on predetermined standards

Unlike a norm-referenced standard, which uses a continuous variable, a criterion – referenced standard is a dichotomy. Terms such as **pass-fail, master-non-mastery** or positive negative are used to describe the dichotomy. A positive test means the presence of the disease and a negative test means the absence of the disease.

Nursing examination is criterion – referenced test. Driving test is also criterion – referenced test. The criterion referenced test is a mastery test, designed to establish how many pupils have achieved a certain standard or whether an individual pupil has performed a given task. Criterion – referenced test reliability examines the consistency of classification. For example, what percentage of people was consistently classified as passing or failing a test that has been administered two times?

Criterion – referenced test validity refers to the accuracy of the classification. That is, are the participants who are classified as passing or failing by the test classified correctly when compared to their true state?

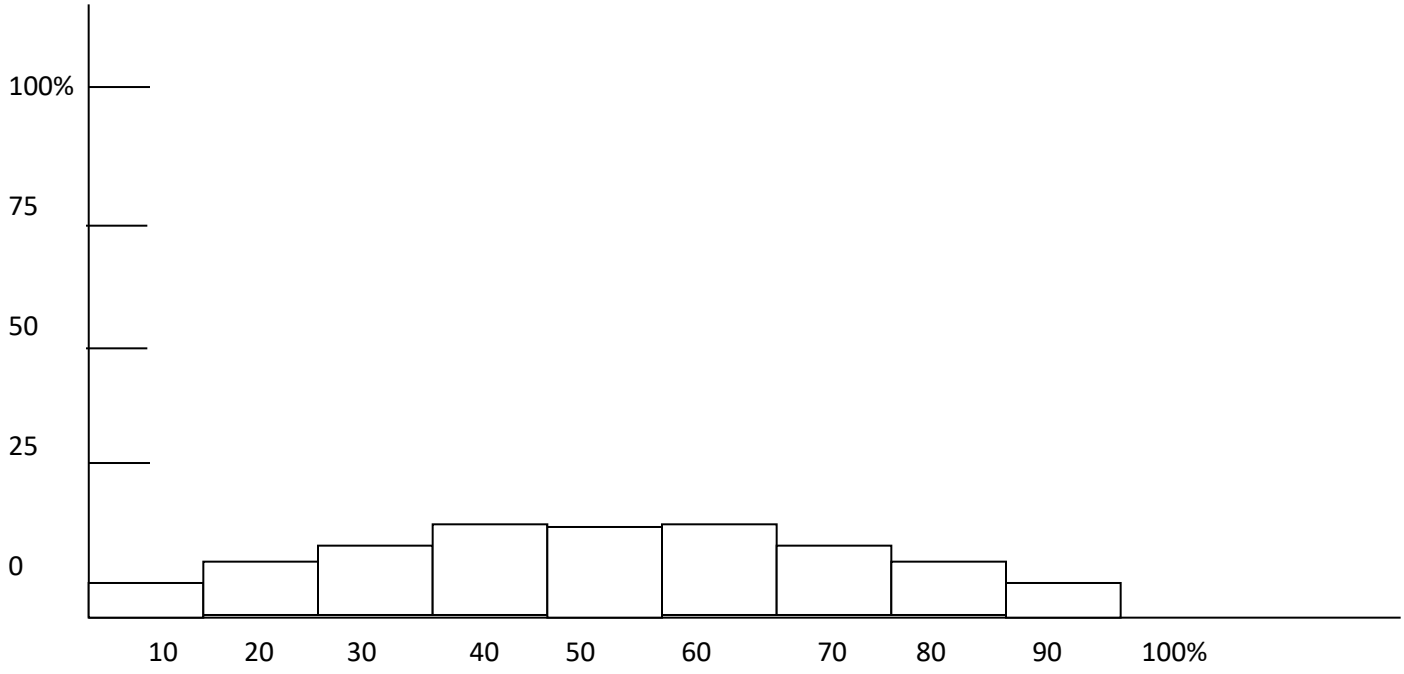
The histograms shown below illustrate the different characteristics of the score distribution in norm and criterion – referenced assessment. The norm referenced assessment has distributed the candidates over the range of marks or grades available while the criterion – referenced assessment has provided a measure of the level of mastery attained by the group.

In summary ***criterion-referenced standard*** involves:

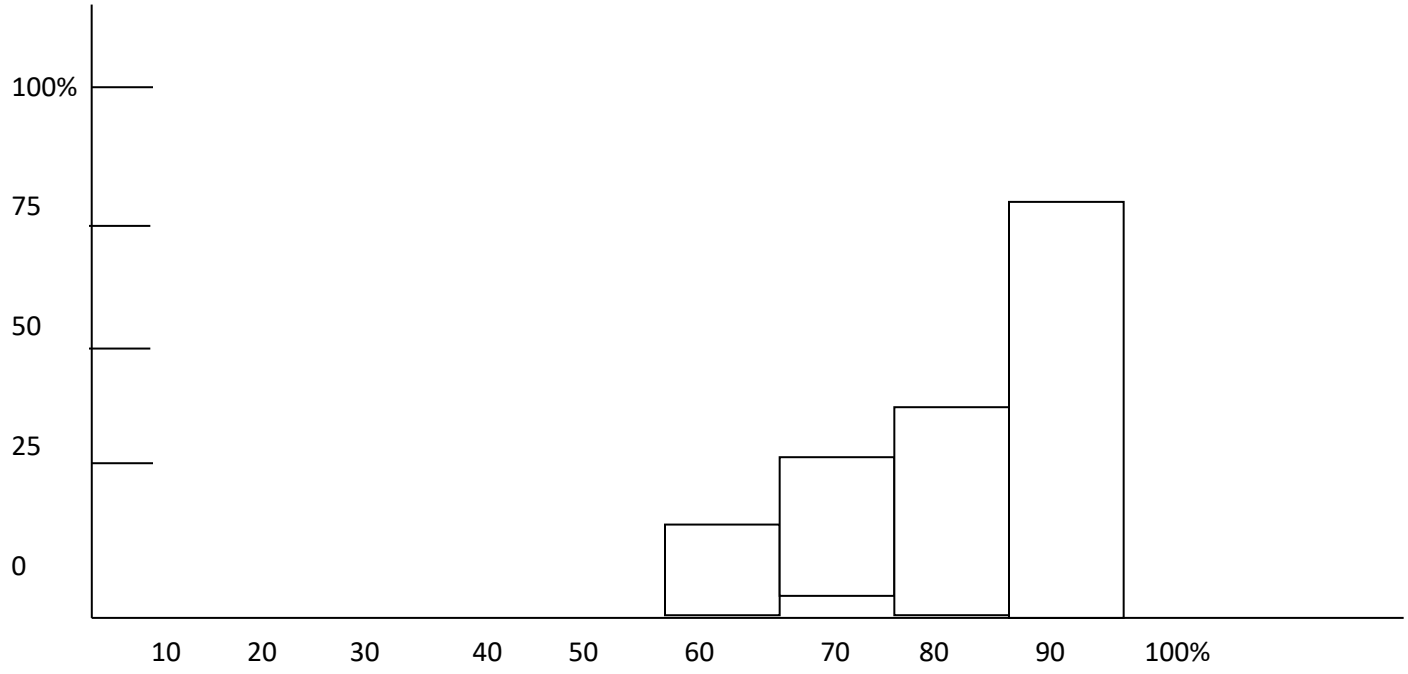
i. Comparing individual's performance with absolute standards- a specification of competence in a given field.
ii. Independent of the group performance. *Performance of the individual is not compared with that of other individuals or the group, but rather just against the standard.*
iii. Interpretation of an individual's score by comparing his/her performance to empirically derived present standards of competence.
iv. Derivation of a normal curve in CRS is unnecessary because discrimination between students is not important

KNUT criticized the wrong use of criterion-referenced standards to grade 2016 KCSE.

**(i) Norm-Referenced**




**Criterion Referenced**



The criterion referenced assessor would be reasonably pleased with the distribution of the scores because his intention was to show whether pupils had learned well enough to attain the objectives. The norm referenced assessor prefers to set his test to ensure a wider spread of scores over a broader band of objectives, such as is necessary for grading results in public examinations. He will be able to tell whether one student is either more or less expert than another, but cannot in the end say what aspect of the subject matter has been mastered or expertise acquired. In other words, he will not be able to say whether this or that objective has been attained.

https://www.youtube.com/watch?v=CDzvkxJnT9A

https://www.youtube.com/watch?v=y0W5uTjxme4

**review questions**

a) Define the term evaluation
b) Explain five functions of evaluation
c) Describe the different types of educational evaluation and their application
d) Describe the standards of evaluation

**Further Reading**

1. Education Measurement and Evaluation : J.Swarupa Rani, Discovery Publishing house.
2. Measurement and Evaluation in Teaching : Norman Edward Gronlund Macmillian
3. Measurement and Assessment in Teaching : Robert L. Linn Pearson Education India.
4. Program Evaluation and performance measurement : James C. Me. David, Laura R.L. Hawthorn, Sage Publications